Ali Shafiee

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alishaffiee.github.io | Codeforces | LinkedIn | IOI Profile | Google Scholar

IOI silver medalist and ICPC World Finalist. PhD researcher in stochastic control, robust optimization, and algorithmic game theory.

Education

Institute of Science and Technology Austria (ISTA) — Vienna, Austria
PhD in Computer Science, grade average 1.0 (Austrian scale, 1.0 highest) — *Sep. 2025 – 2028 (expected)*

Sharif University of Technology — Tehran, Iran
BSc in Computer Engineering, GPA 3.68/4 (17.03/20) — *Sep. 2018 – Aug. 2023*

Publications

Author lists are alphabetical, per convention in theoretical computer science.

- **On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets**. A. Asadi, K. Chatterjee, A. Montaseri, **A. Shafiee**. *NeurIPS 2026, spotlight.* [paper]
- **Randomise Alone, Reach as a Team**. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026.* [paper]
- **Qualitative Analysis of ω-Regular Objectives on Robust MDPs**. A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026.* [paper]
- **Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete**. A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025.* [paper]
- **Quantitative Analysis of ω-Regular Robust MDPs**. A. Asadi, K. Chatterjee, E. K. Goharshady, M. Karrabi, A. Montaseri, **A. Shafiee**. *AAAI 2027, under review.* [paper]
- **Limit-sure Winning in Team Games without Shared Randomness**. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *AAMAS 2027, under review.* [paper]
- **Mechanism Design for Multi-Resource Nakamoto Consensus**. N. Babashah, M. A. Baig, K. Pietrzak, **A. Shafiee**. *FC 2027, under review.*

Awards

- **Codeforces**: **top 50 worldwide in 2019**; ex-International Grandmaster, peak rating 2800 (handle `ToTLeS`)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

Industry Experience

Quera, Iran's largest competitive-programming and technical-education platform — Tehran, Iran

Business Line Manager, Quera College — *Oct. 2024 – Aug. 2025*
- Owned Quera College end to end: strategy, roadmap, delivery and OKR planning.
- Grew revenue by **68%** and active users by **49%** over one year, and delivered **200+ hires** to corporate partners.

Director of Computer Science Education — *Nov. 2022 – Jun. 2024*
- Led the education department and set the strategic roadmap for its products, while overseeing core company operations.
- Supervised and mentored six cross-functional technical teams, a division of roughly **40 engineers, content creators and specialists**, spanning back-end, front-end, data, blockchain, algorithms and fundamentals.
- Streamlined the content delivery pipeline, introduced agile practice across teams, and scaled the platform's infrastructure for large-scale technical training.

Technical Content and Education Manager — *Mar. 2020 – Oct. 2022*
- Designed and shipped **12+ industry-oriented bootcamps and online courses** at Quera College, covering C++, Python, SQL, Django, Linux, Git, MATLAB, data analysis, machine learning and deep learning.
- Managed several curriculum teams, holding material to a standard of technical accuracy and keeping it aligned with current industry practice.

Manager of Quera Contest — *Mar. 2019 – Mar. 2020*
- Organised **20+ large-scale programming contests** in algorithms, back-end, front-end and machine learning, and grew an active competitive-programming community around them.
- Ran sponsored hiring hackathons with corporate partners as a technical recruiting channel.

Rahnema College — Tehran, Iran

Software Engineering Intern — *Jun. 2018 – Sep. 2018*
- Contributed to the back-end design and development of a highly interactive social media application.

Skills

Programming: C++, Python, SQL, Rust · **Numerical & ML**: NumPy, SciPy, pandas, scikit-learn, PyTorch, JAX
Mathematics: probability, stochastic processes, Markov decision processes, optimization, game theory
Tools: Git, Linux/Bash, Docker, LaTeX · **Languages**: Persian (native), English (fluent), German (basic)

Research Experience

Institute of Science and Technology Austria (ISTA) Vienna, Austria
Research Intern and Online Collaborator, then PhD Researcher, advisor Prof. Krishnendu Chatterjee *Apr. 2024 – present*

- Proved that limit-sure reachability for small-memory POMDP policies is **NP-complete**, settling a case whose unrestricted version is undecidable (UAI 2025).
- Established the complexity of robust MDPs under ω-regular objectives: oracle-based algorithms for the qualitative problem with no structural assumptions (AAAI 2026), and a polynomial-time algorithm for quantitative parity analysis on robust Markov chains (AAAI 2027, under review).
- Showed policy iteration is **strongly polynomial** for discounted robust MDPs with L_1 and L_∞ uncertainty sets, and proved hardness for the intermediate L_p norms (**NeurIPS 2026 spotlight**).
- Analysed concurrent team games in which players randomise independently: memoryless strategies suffice, and limit-sure reachability is NP-complete (CAV 2026; AAMAS 2027, under review).
- Implemented and benchmarked the solvers against existing approaches: on robust-MDP reachability the implementation cleared **426 benchmark instances** to the prior baseline's **81**, averaging **0.1 s** against over **6 s** on instances both solved, over models of up to 10^4 states (AAAI 2026).
- On the Garnet and inventory-management benchmarks, the quantitative parity solver ran **one to two orders of magnitude** faster than the stochastic-game reduction, which times out at a few hundred states (AAAI 2027).

Sharif University of Technology Tehran, Iran
Research Assistant, advisor Prof. Mohammad Ali Abam *Jun. 2023 – Aug. 2023*

- Studied round- and memory-efficient algorithms for minimum cut and maximal matching in the Massive Parallel Computation (MPC) and Adaptive MPC models.

Teaching & Service

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, IOI 2019, Baku
- **Problem-setter** (2017 – present) for the Iranian National Olympiad in Informatics, Codecup and Quera; **olympiad coach** (2017 – 2023), 50+ students trained for INOI and the IOI
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming
- **Course author**, Quera College: co-designed eight published programming courses, including Python (introductory and advanced), Front-End, Django, Git, C and C++, with **46,000+ enrolled learners**
- **Conference talks**: presented at UAI 2025 and CAV 2026; NeurIPS 2026 upcoming (December 2026)